Statistical Inference and Multivariate Analysis (MA324)

LECTURE SLIDES

Topic 03

Multiple Linear Regression

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Multiple Linear Regression

- In general, the response (y) may be related to p regressors (input variables/predictors).
- The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip} + \epsilon_i, i = 1, \ldots, n$$

  is called **multiple linear regression**. A regression model with p regressors.
- The parameters $\beta_j, j = 0, 1, 2, \cdots, p$ are called regression coefficients.
- $\beta_j, j = 0, 1, 2, \cdots, p$ represents the change in the average value of the response for a unit change in j^{th} regressor keeping other regressors fixed.

- Data (of sample size n): $(y_i, x_{i1}, x_{i2}, \cdots, x_{ip}), i = 1, \ldots, n$
- Scalar notation becomes cumbersome – so matrix notation is used
- Let

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \; X = \begin{pmatrix} 1 & x_{11} & \ldots & x_{1p} \\ \vdots & \vdots & x_{ij} & \vdots \\ 1 & x_{n1} & \ldots & x_{np} \end{pmatrix}, \; \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \; \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

- Then we can write the model in a more compact form:

$$y_{n \times 1} = X_{n \times (p+1)} \beta_{(p+1) \times 1} + \epsilon_{n \times 1}$$

- X is called the ***design matrix***

Error Assumptions

- As before, ϵ_i's satisfy
  - $E(\epsilon_i) = 0$ for $i = 1, \ldots, n$
  - $Var(\epsilon_i) = \sigma^2$ for $i = 1, \ldots, n$
  - $Cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$
- Equivalently, the assumptions can be written as
  - $E\left(\boldsymbol{\epsilon}\right) = \mathbf{0}$
  - $Var(\boldsymbol{\epsilon}) = \sigma^2 I_n$

Estimation of Regression Coefficients:

- Consider the objective function

$$\begin{aligned} Q(\boldsymbol{\beta}) &= Q(\beta_0, \beta_1, \cdots, \beta_p) \\ &= \sum_{i=1}^{n} \Big(y_i - \beta_0 - \beta_1 x_{i1} - \ldots - \beta_p x_{ip} \Big)^2 \\ &= (y - X\beta)^T (y - X\beta) \\ &= \left(y^T y - 2\beta^T X^T y + \beta^T X^T X \beta \right) \end{aligned}$$

- The LSEs of $\beta_0, \beta_1, \cdots, \beta_p$ are $\underset{\boldsymbol{\beta}}{\operatorname{argmin}}\, Q(\boldsymbol{\beta})$.

Vector Derivative : Linear Function

- Let $\boldsymbol{x}$ be an $n \times 1$ column vector (the variable).
- Let $\boldsymbol{a}$ be a constant $n \times 1$ column vector.
- The derivative of a scalar-valued linear function with respect to the vector $\boldsymbol{x}$ is
$$\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{a}^T \boldsymbol{x}) = \boldsymbol{a}.$$

Vector Derivative: Quadratic Function

- Let A be a constant $n \times n$ matrix.
- The derivative of a quadratic form with respect to the vector x is

$$\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}^T A\boldsymbol{x}) = \left(A + A^T\right)\boldsymbol{x}.$$

- If A is a symmetric matrix, then the above formula simplifies to

$$\frac{\partial}{\partial \boldsymbol{x}}\left(\boldsymbol{x}^T A\boldsymbol{x}\right) = 2A\boldsymbol{x}.$$

Estimation of Model Parameters:

- Differentiating $Q(\beta)$ with respect to β and setting the derivative to zero gives the following *normal equations*:

$$X^T X\beta = X^T y.$$

- The above system of linear equations in $\boldsymbol{\beta}$ is always consistent.

- If the matrix $X^T X$ is *invertible* (i.e. if X is of *full rank*), then the LSE is unique and given by

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Fitted Value:

- The fitted value of response corresponding to regression $x = (1, x_1, \cdots, x_p)$ is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \ldots + \hat{\beta}_p x_p$$

- Then $\hat{y} = (\hat{y}_1, \hat{y}_2, \cdots, \hat{y}_n)^T = X\hat{\beta} = X(X^T X)^{-1} X^T y = Hy$, where, $H = X(X^T X)^{-1} X^T$ is called **hat-matrix.**

- The residuals:

$$\boldsymbol{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (I - H)\boldsymbol{y}$$

Properties of LSE of β

- $\hat{\beta}$ is a linear function of y
- $\hat{\beta}$ is unbiased estimator of β. That is, $E(\hat{\beta}) = \beta$
- $Var\left(\hat{\beta}\right) = \sigma^2 (X^T X)^{-1}$
- $\hat{\beta}$ is the BLUE of β.

Estimation of Error Variance (σ^2)

- It can be shown that

$$E(SS_{Res}) = (n - p - 1)\sigma^2$$

- Hence, $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1} = MS_{Res}$ is an unbiased estimator of σ^2. Here MS_{Res} is residual mean square.
- Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1}$ is called ***Residual variance***. It's **square root is called *residual standard error*.**
- $\frac{(n-p-1)MS_{Res}}{\sigma^2} \sim \chi^2_{n-p-1}$.
- Here,

$$SS_{Res} = \sum_{i=1}^{n} e_i^2 = y^T y - \hat{\beta}_1^T X^T y,$$

Hypothesis Testing: Test for Significance of Regression

- Assumption: $\epsilon \sim N_n(0, \sigma^2 I_n)$.
- Want to test the hypothesis if there is a linear relationship between the response y and any of the regressors $x_1, \cdots, x_n$:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \text{ ag. } H_1 : \beta_j \neq 0 \text{ for atleast one } j = 1, \cdots, p$$

- The test statistic is

$$F_0 = \frac{SS_{Reg}/p}{\hat{\sigma}^2} = \frac{SS_{Reg}/p}{SS_{Res}/(n-p-1)} \sim F_{p,n-p-1}, \text{ under } H_0.$$

- Reject H_0 iff $F_0 > F_{p,n-p-1;\alpha}$ (at level α)

Hypothesis Testing: Significance of regression coefficients

- Want to test

$$H_0 : \beta_j = 0 \text{ ag. } H_1 : \beta_j \neq 0$$

- Note that

$$\hat{\beta} \sim N_p\left(\beta, \sigma^2 \left(X^T X\right)^{-1}\right) \implies \hat{\beta}_j \sim N\left(\beta_j, \sigma^2 C_{jj}\right),$$

where C_{jj} is the j-th diagonal element of $(X^T X)^{-1}$

- The test statistic is

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{MS_{Res} C_{jj}}} \sim t_{n-p-1} \text{ under } H_0$$

- Reject H_0 iff $|t_0| > t_{n-p-1,\alpha/2}$ (at level α)

Test of contribution of a subset of the regressors

- $$y = X\beta + \epsilon \implies y = (X_1, X_2)\begin{pmatrix}\beta_1 \\ \beta_2\end{pmatrix} + \epsilon$$
- Want to test:
$$H_0 : \beta_2 = 0 \text{ ag. } H_1 : \beta_2 \neq 0$$
- Based on the full model ($y = X\beta + \epsilon$), $\hat{\beta} = (X^T X)^{-1} X^T y$ and $SS_{Reg}(\beta)$ has p degrees of freedom.
- To find the contribution of β_2, fit the model assuming $\beta_2 = 0$.
- The reduced model is $y = X_1\beta_1 + \epsilon$, $\hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T y$ and $SS_{Reg}(\beta_1)$ has $p - r$ degrees of freedom. Where r denotes the number of components in β_2

Test of contribution of a subset of the regressors

- $SS_{Reg}(\beta_2 \mid \beta_1) = SS_{Reg}(\beta) - SS_{Reg}(\beta_1)$ can be used as a measure of contribution of β_2

- Note that if β_2 has significant contribution then $SS_{Reg}(\beta_2 \mid \beta_1)$ is large

- Therefore, the test statistic is

$$F_0 = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{\hat{\sigma}^2} = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{SS_{Res}/(n-p-1)} \sim F_{r,n-p-1}, \text{ under } H_0$$

- Reject H_0 iff $F_0 > F_{r,n-p-1;\alpha}$ (at level α)

Confidence intervals for β_j:

- Confidence Interval of individual regression coefficient $\hat{\beta}_j$
- Pivot is

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{MS_{Res} C_{jj}}} \sim t_{n-p-1},$$

where C_{jj} is the diagonal element of $(X^T X)^{-1}$ matrix.

- A $100(1-\alpha)\%$ CI for β_j is

$$\left[\hat{\beta}_j \pm t_{n-p-1,\alpha/2} \sqrt{MS_{Res} C_{jj}} \right].$$

Confidence interval for mean response:

- Let $x_0 = \begin{pmatrix} 1 \\ x_{01} \\ \vdots \\ x_{0p} \end{pmatrix}$ be a value of the regressor vector
- The mean response at x_0 is ${x_0}^T\beta$
- Pivot is $\frac{\hat{y}_0 - {x_0}^T\beta}{\sqrt{MS_{Res}{x_0}^T(X^TX)^{-1}x_0}} \sim t_{n-p-1}$
- A $100(1-\alpha)\%$ CI for ${x_0}^T\beta$ is

$$\left[\hat{y_0} \pm t_{n-p-1,\alpha/2}\sqrt{MS_{Res}{x_0}^T(X^TX)^{-1}x_0}\right]$$

Prediction Interval for New Observation

- Consider a level of regressor x_0.
- Let y_0 be the corresponding value of the response.
- We want a prediction interval for y_0.
- Let $\Psi = y_0 - \hat{y_0} = y_0 - {x_0}^T\hat{\beta} \sim N(0, \sigma^2(1 + {x_0}^T(X'X)^{-1}x_0))$.
- A pivot is $\frac{\Psi - 0}{\sqrt{MS_{Res}(1 + {x_0}^T(X'X)^{-1}x_0)}} \sim t_{n-p-1}$.
- A $100(1-\alpha)\%$ prediction interval of y_0 is

$$\left[\hat{y_0} \mp t_{n-p-1;\frac{\alpha}{2}}\sqrt{MS_{Res}(1 + {x_0}^T(X'X)^{-1}x_0)}\right]$$

Standardized Regression Coefficients

- Difficult to compare regression coefficients. The magnitude of β_j reflects the unit of measurement of regressor x_j.

- For example, $y = 5 + x_1 + 1000x_2$, where y is measured in liters, x_1 in milliliters, and x_2 in liters. Here, $\beta_2 >> \beta_1$. But the effects of both regressor on y are identical.

- Wayout is to standardized the regressors and response so that they become unit free.

Standardized Regression Coefficients

- A popular approach is as follows:
  - Define $W_{ij} = \frac{x_{ij} - \overline{x}_j}{\sqrt{S_{jj}}}$, $i = 1, 2, ..., n; j = 1, 2, ..., p.$
    $y_i^* = \frac{y_i - \overline{y}}{\sqrt{SS_T}}$, $i = 1, 2, ..., n.$
    Here $S_{jj} = \sum_{i=1}^{n}(x_{ij} - \overline{x_j})^2$, $j = 1, 2, ..., p.$
  - $\overline{W}_j = 0$ and $\sqrt{\sum_{i=1}^{n}(W_{ij} - \overline{W}_j)^2} = \sqrt{\sum_{i=1}^{n} W_{ij}^2} = 1$
    $\overline{y}^* = 0$ and $\sqrt{\sum_{i=1}^{n}(y_i^*)^2} = 1$
  - In terms of $y^*, W_1, ..., W_p$, the regression model becomes,

$$y^* = b_1 W_1 + b_2 W_2 + ... + b_p W_p + \epsilon$$

  - In matrix notation: $y^* = Wb + \epsilon$.
  - LSE: $\hat{b} = (W^{'}W)^{-1}W^{'}y^*$.

Standardized Regression Coefficients

- $W'W = \begin{pmatrix} 1 & r_{12} & . & . & . & r_{1p} \\ r_{21} & 1 & . & . & . & r_{2p} \\ . & . & & & & . \\ . & . & & & & . \\ . & . & & & & . \\ r_{p1} & r_{p2} & . & . & . & 1 \end{pmatrix}_{p\times p} \quad W'y^* = \begin{pmatrix} r_{1y} \\ r_{2y} \\ . \\ . \\ . \\ r_{py} \end{pmatrix}_{p\times 1}$
- $r_{ij} = \frac{\sum_{u=1}^{n}(x_{ui}-\bar{x}_i)(x_{uj}-\bar{x}_j)}{\sqrt{S_{jj}S_{ii}}} = \frac{S_{ij}}{\sqrt{S_{jj}S_{ii}}}$
- $r_{iy} = \frac{\sum_{u=1}^{n}(x_{ui}-\bar{x}_i)(y_u-\bar{y})}{\sqrt{S_{jj}S_{ii}}} = \frac{S_{jy}}{\sqrt{S_{jj}SS_T}}$

Model Adequacy Checking (Regression Diagnostics)

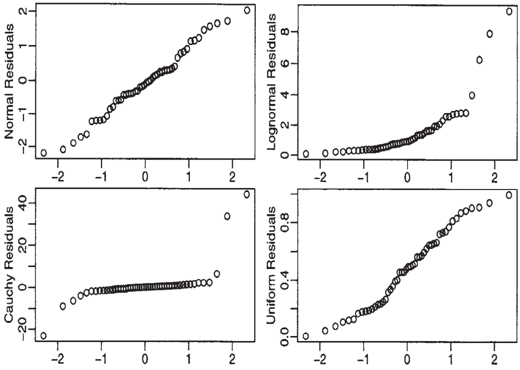
- Major assumptions
  - linear relationship
  - Error mean zero
  - Error variance is constant
  - Error are uncorrelated
  - Error are normally distributed
- Gross violation of the assumptions may lead to a totally different model with opposite conclusions.
- Unusual observations: Sometimes just a few observations do not fit the model. These few observations might change the choice and fit of the model.
- We perform the checking using residuals.

Different Residuals

- We now define 3 types of residuals.
- Residual: $e_i = y_i - \hat{y}_i$.
- Standardized residual: $d_i = \frac{e_i}{\sqrt{MS_{Res}}}$
- Studentized residual:
  - $e = (I - H)\epsilon \Rightarrow Var(e) = \sigma^2(I - H)$.
  - $Var(e_i) = \sigma^2(1 - h_{ii})$ and $Cov(e_i, e_j) = -\sigma^2 h_{ij}$.
  - It can be shown that $0 \leq h_{ii} \leq 1$.
  - $r_i = \frac{e_i}{\sqrt{MS_{Res}(1-h_{ii})}}$.

Residual Plots: Q-Q Plot (Test for normality)

- The residuals can be assessed for normality using a Q–Q plot. This compares the residuals to “ideal” normal observations.
- We plot the quantiles corresponding to sorted residuals (e_i) against $\Phi^{-1}(\frac{i}{n+1})$ for $i = 1, \cdots, n$.
- Four Q-Q plots are:
  Normal: ideal;
  Lognormal: an example of a skewed distribution;
  Cauchy: an example of a long-tailed (platykurtic) distribution;



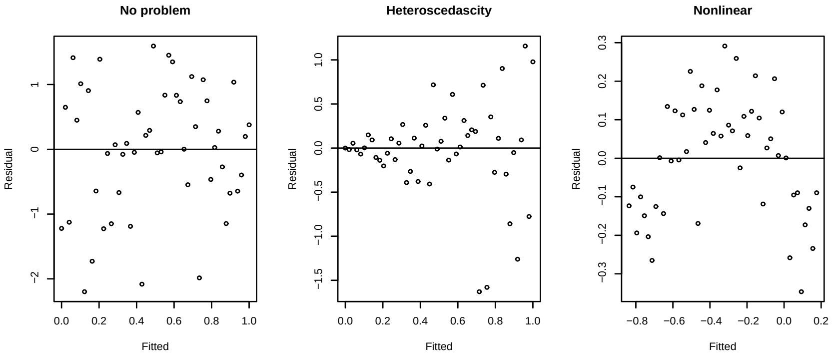
- **Uniform:** an example of a short-tailed (leptokurtic) distribution
- Source (book): Linear Models with R by Julian J.Faraway

If non-Normal?

- For short-tailed distributions, the consequences of nonnormality are not serious and can reasonably be ignored.
- For skewed errors, a transformation of the response may solve the problem.
- For heavy-tailed errors, we might just accept the nonnormality and base the inference on the assumption of another distribution or use resampling methods such as the bootstrap.

Residual Plots: Plot of residual against Fitted values

- Plot $\hat{y}_i$ vs $e_i(or\ d_i\ or\ r_i)$: (Test of constant variance and non-linear relation)



Residuals vs Fitted plots - the **first** suggests **no change** to the current model while the **second** shows **non-constant variance** and the **third** indicates **some nonlinearity** which should prompt some change in the structural form of the model

Residual Plots

- Plot of residual against regressor
  - Plot x_{ij} vs $e_i \forall j$
  - In previous plot, replace 'fitted' by x_{ij} to find similar interpretation.
  - Repeat it for all the regressors.

If nonconstant variance?

- Two approaches
  - Weighted least squares
  - Transformation of variables
- Let h be a transformation that is applied on response

$$h(y) = h(Ey) + (y - Ey)h'(Ey) + \cdots$$
$$\text{var } h(y) = h'(Ey)^2 \text{var } y + \cdots$$

We ignore the higher order terms. For var $h(y)$ to be constant we need

$$h'(Ey) \propto (\text{var } y)^{-1/2},$$

which suggests

$$h(y) = \int \frac{dy}{\sqrt{\text{var } y}} == \int \frac{dy}{\text{SDy}}.$$

- If var $y =$ var $\varepsilon \propto (Ey)^2$, then $h(y) = \log y$ is suggested.
- If var $\varepsilon \propto (Ey)$, then $h(y) = \sqrt{y}$.

Checking Correlated Errors

- Index plot: Plot residual against index (time).
- Plot e_i against e_{i-1}.
- Remedy: Use generalized least squares method

Checking the Structure of the Model

- How do we check whether the systematic part $(Ey = X\beta)$ of the model is correct?
- Partial Regression Plot:
  - Complete/correct marginal effect
  - Let we want to study the marginal effect of x_j on y
  - y is regressed on $x_1, ..., x_k$ except x_j.
  - x_j is regressed on $x_1, ..., x_k$ except x_j.
  - Plot y residual, $e_i(y|x_1, ..., x_{j-1}, x_{j+1}, ..., x_k)$ against x_j residual, $e_i(x_j|x_1, ..., x_{j-1}, x_{j+1}, ..., x_k)$
  - For ideal scenario, the partial regression plot should show a linear relationship (straight line with non-zero slope).
  - Curvilinear band: indicates higher order terms in x_j or it's transformation.
  - Horizontal band: indicates no additional useful information in x_j
- Remedy: Change you model and/or add other regressors.